HDOC-17

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Program to Check if a Number is an Armstrong Number

This document provides a clean, well-commented implementation in C that checks whether an integer is an Armstrong number. An Armstrong number (narcissistic number) is a number that equals the sum of its own digits, each raised to the power of the number of digits. For example, $153 = 1^3 + 5^3 + 3^3$.

1. C Implementation

Step 1: Understand the Logic

First count the digits (n) of the number. Then extract each digit using (num % 10), raise it to the power n, and add it to a running sum. Finally compare the sum with the original number. If they are equal, it is an Armstrong number.

- **Digit count:** loop with temp /= 10 until temp becomes 0.
- **Sum:** sum += $digit^n$ for every digit.
- **Decision:** if sum == originalNum, print Armstrong; otherwise, not Armstrong.

Step 2: Define Variables and Data Types

Use int for the input (num), saved copy (originalNum), digit (remainder), digit count (digits) and loop counter (i). Use long long for power and sum so that larger values do not overflow. The power is computed with a multiplication loop instead of pow(), which avoids floating-point rounding errors.

Step 3: Take User Input

Prompt the user using printf("Enter a positive integer: "); and read using scanf("%d", &num);.

Step 4: Count, Sum and Compare

1. Store the input in originalNum.
2. Count digits using a while loop.
3. In a second while loop, extract each digit, raise it to the power digits using a for loop, and add to sum.
4. Use if-else to compare sum with originalNum and print the result.

Complete C Code

```
/*
 * Q28 (User Inputs, Operations & Output)
 * Write a program to check if a number is an Armstrong number.
 */
#include <stdio.h>

int main() {
    int num, originalNum, remainder, digits = 0, i;
    long long power, sum = 0;

    // Step 1: Read the value of num
    printf("Enter a positive integer: ");
    scanf("%d", &num);
    originalNum = num;              // Save original for later use

    // Step 2: Count the number of digits
    int temp = num;
    while (temp != 0) {
        digits++;
        temp /= 10;
    }

    // Step 3: Add each digit raised to the power of 'digits'
    temp = num;
    while (temp != 0) {
        remainder = temp % 10;
        power = 1;
        for (i = 0; i < digits; i++) {
            power *= remainder;
        }
        sum += power;
        temp /= 10;
    }

    // Step 4: Compare the sum with the original number
    if (sum == originalNum)
        printf("%d is an Armstrong number.\n", originalNum);
    else
        printf("%d is not an Armstrong number.\n", originalNum);

    return 0;
}
```

Program Output (Sample Runs)

```
Run 1:  $ gcc hdoc17.c -o hdoc17
        $ ./hdoc17
        Enter a positive integer: 153
        153 is an Armstrong number.

Run 2:  $ ./hdoc17
        Enter a positive integer: 9474
        9474 is an Armstrong number.

Run 3:  $ ./hdoc17
        Enter a positive integer: 123
        123 is not an Armstrong number.

Run 4:  $ ./hdoc17
        Enter a positive integer: 10
        10 is not an Armstrong number.

Run 5:  $ ./hdoc17
        Enter a positive integer: 5
        5 is an Armstrong number.
```

Dry Run with Test Cases

Sample Test Case 1

Input: num = 153 (originalNum = 153)
Calculation:
Digit count: 153 $\rightarrow$ 15 $\rightarrow$ 1 $\rightarrow$ 0, so digits = 3.
Pass 1: remainder = 3, power = 3^3 = 27, sum = 27, temp = 15.
Pass 2: remainder = 5, power = 5^3 = 125, sum = 152, temp = 1.
Pass 3: remainder = 1, power = 1^3 = 1, sum = 153, temp = 0.
Compare: sum (153) == originalNum (153) is TRUE.
Output: 153 is an Armstrong number.

Sample Test Case 2

Input: num = 123 (originalNum = 123)
Calculation:
digits = 3.
Pass 1: remainder = 3, power = 27, sum = 27.
Pass 2: remainder = 2, power = 8, sum = 35.
Pass 3: remainder = 1, power = 1, sum = 36.
Compare: sum (36) == originalNum (123) is FALSE.
Output: 123 is not an Armstrong number.

Sample Test Case 3

Input: num = 5
Calculation: digits = 1. Pass 1: remainder = 5, power = 5^1 = 5, sum = 5. Compare: 5 == 5 is TRUE.
Output: 5 is an Armstrong number.